ModelIt!

Teacher's Guide

Ecohydrology — Fire-Resistant Landscapes

How Beavers Create Nature's Firebreaks

Quick Reference

Grade Level	Grade NE
NGSS Standard	MS-ESS2-4, MS-ESS3-3, MS-ESS3-4, MS-LS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS2-4, MS-ESS3-3, MS-ESS3-4, MS-LS2-4

Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity. Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop an 8-component computational model showing how beaver-driven water management creates fire-resistant green corridors through cascading ecohydrological relationships.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Surface Processes** — Water movement through beaver-dammed landscapes raises groundwater, maintains soil moisture, and sustains vegetation — demonstrating how water cycling shapes surface conditions and fire behavior.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from beaver dams through groundwater, soil moisture, and vegetation health to fire spread rate — identifying that each step in the cascade has a predictable effect on the next.

CAST Testing Alignment

MULTIPLE CHOICE:

During a severe drought, satellite imagery shows a narrow strip of green vegetation along a stream while the surrounding hillsides are brown and dry. According to the ecohydrology model, what is the MOST LIKELY explanation? A) The stream has more rainfall than the hillsides B) Beaver dams raised the groundwater level, keeping soil moist and vegetation green C) The vegetation along the stream is a different species that does not need water D) The fire burned everything except the stream area by coincidence

CONSTRUCTED RESPONSE:

Using your 8-component model, explain step by step why a landscape WITHOUT beaver dams is more vulnerable to wildfire during drought than one WITH beaver dams. Trace the cascade from Beaver Dam Count through Groundwater Level, Soil Moisture, Vegetation Health, and Fire Spread Rate. Use evidence from your simulation scenarios.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson builds on Level 1 Sponge vs. Pavement by expanding the water absorption concept into a full ecohydrology system. Students now see how beaver-engineered water management creates fire-resistant landscapes — based on peer-reviewed research by Dr. Emily Fairfax and colleagues.

The Fairfax Research

In 2020, Dr. Emily Fairfax and colleagues published groundbreaking research in the journal Ecosphere showing that beaver-dammed sections of streams maintained green, hydrated vegetation corridors even when surrounding landscapes were scorched by wildfire. Using satellite imagery from multiple large wildfires across the American West, the research team demonstrated that beaver-created wetlands acted as natural firebreaks. The key mechanism: beaver dams raise the water table, keeping soil saturated and vegetation too moist to burn.

How Beaver Dams Manage Water

Beaver dams do not just create ponds — they fundamentally change how water moves through a landscape. By slowing streamflow, beaver dams force water to spread laterally across the floodplain. This water soaks into the ground, raising the water table for hundreds of meters on either side of the stream. The elevated water table keeps plant roots in contact with moisture even during surface-level drought. The result: a green, hydrated ribbon of vegetation surrounded by dry, fire-prone land.

Fire Behavior and Vegetation Moisture

Fire spreads through dry fuel — dead leaves, dry grass, and stressed vegetation with low moisture content. When vegetation is well-hydrated, it resists ignition and slows fire spread dramatically. A strip of green, moist vegetation can act as a natural firebreak, stopping or redirecting a wildfire the same way a paved road or a cleared strip would. The difference is that beaver-created firebreaks are LIVING ecosystems that also support wildlife, filter water, and store carbon.

Beavers as Climate Adaptation

As climate change increases drought frequency and wildfire intensity across the western United States, land managers are increasingly looking to beavers as a low-cost, self-maintaining wildfire resilience strategy. Reintroducing beavers to degraded streams costs far less than engineered firebreaks and provides year-round ecosystem services. Several states have begun 'beaver relocation' programs, moving conflict beavers from urban areas to fire-prone watersheds.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Precipitation, Beaver Dam Count, and Drought Severity as external components — environmental conditions that can be observed and set. Groundwater Level, Soil Moisture, Vegetation Health, Surface Runoff, and Fire Spread Rate are internal components that change based on the externals and each other.

E **Establish Relationships**

Students trace the relationships: Beaver Dam Count positively affects Groundwater Level and negatively affects Surface Runoff. Precipitation positively affects Groundwater Level. Drought Severity negatively affects Soil Moisture. Groundwater Level positively affects Soil Moisture. Soil Moisture positively affects Vegetation Health. Vegetation Health negatively affects Fire Spread Rate.

V **Visualize & Model**

Students build the full 8-component model with three external inputs feeding five internal components through a web of positive and negative relationships.

E **Evaluate Outcomes**

Students run four scenarios — with beavers, without beavers, normal conditions, and extreme drought — to see how beaver presence dramatically changes fire outcomes.

R **Revise & Extend**

Students add Wind Speed, Terrain Slope, or Vegetation Density to explore how additional factors modify the fire behavior equation.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at this image. The land burned everywhere EXCEPT along this stream. Why? Today we figure out how beavers create nature's firebreaks."*

Watch For

Show satellite or aerial imagery from Dr. Fairfax's research. Ask: What do you notice about the green area vs. the burned area?

Slide 2: Learning Objectives

3 min

Say: *"We are building an 8-component ecohydrology model — our biggest yet. This model explains how water underground connects to fire above ground."*

Watch For

Preview vocabulary: ecohydrology, firebreak, groundwater, green corridor. Connect to Level 1 Sponge vs. Pavement knowledge.

Slide 3: Big Question

3 min

Say: *"Beavers do not spray water on fires. They do not clear brush. So how do scientists say they create fire-resistant landscapes?"*

Watch For

Think-pair-share: How could an animal that builds dams in water affect fire on land? Students brainstorm connections.

Slide 4: LEVER Framework

2 min

Say: *"Our model today has BOTH positive and negative relationships — and the cascade goes from water underground to fire above ground."*

Watch For

Preview the cascade direction: water inputs → groundwater → soil → vegetation → fire. Emphasize underground-to-surface connection.

Slide 5: Activity 1: Components

8 min

Say: *"Three things the environment controls, five things the SYSTEM produces. Let us sort them — which are inputs and which are outputs?"*

Watch For

Guide sorting of all 8 components. Key discussion: Drought Severity is external because humans do not control it, but it has powerful effects on the system.

Slide 6: Activity 2: Connections

10 min

Say: *"Here is where it gets fascinating. Beaver dams push Groundwater UP but push Surface Runoff DOWN. And Vegetation Health pushes Fire Spread Rate DOWN. These negative relationships are KEY."*

Watch For

Build the relationship web arrow by arrow. Color-code: green arrows for positive, red arrows for negative. Key teaching moment: the negative relationship between Vegetation Health and Fire Spread Rate.

Slide 7: Activity 3: Simulation

12 min

Say: *"Our big test: same drought, same landscape — the ONLY difference is beaver dams. Predict what happens, then watch the model."*

Watch For

Run With Beavers vs. Without Beavers scenarios back-to-back. Record predictions BEFORE running. The contrast is dramatic. Then test Normal and Extreme Drought.

Slide 8: Discoveries

4 min

Say: *"Beavers are not firefighters — they are water managers. And managing water turns out to be the best way to manage fire."*

Watch For

Summarize key findings. Show Dr. Fairfax's real satellite imagery. Discuss: How could this change wildfire policy?

Slide 9: STEM Challenge

12 min

Say: *"Now build it and prove it. Two terrariums, two weeks, one question: does the dam keep the landscape green?"*

Watch For

Distribute terrarium materials. Groups set up paired terrariums with and without dams. Begin the two-week drought observation.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Precipitation, Beaver Dam Count, Drought Severity
- **INTERNAL:** Groundwater Level, Soil Moisture, Vegetation Health, Surface Runoff, Fire Spread Rate

Relationships

- **Beaver Dam Count to Groundwater Level:** POSITIVE (+) — More beaver dams slow water flow and spread it across the floodplain, forcing water to soak underground. Each dam raises the water table in the surrounding area by creating sustained water contact with the soil.
- **Beaver Dam Count to Surface Runoff:** NEGATIVE (-) — More beaver dams trap and slow surface water, reducing the amount that flows downstream as runoff. Water that would have rushed away is instead stored in ponds and soaked into the ground.
- **Precipitation to Groundwater Level:** POSITIVE (+) — More rain and snow provide more water to recharge groundwater. Precipitation is the primary water input — without it, even beaver dams cannot maintain groundwater indefinitely.
- **Drought Severity to Soil Moisture:** NEGATIVE (-) — Longer, more intense drought dries out the upper soil layers through evaporation and plant water use. Drought directly attacks soil moisture from the surface downward.
- **Groundwater Level to Soil Moisture:** POSITIVE (+) — Higher groundwater tables keep the lower and middle soil layers saturated, which wicks moisture upward to plant roots. Even when the surface is dry, elevated groundwater maintains deeper soil moisture.
- **Soil Moisture to Vegetation Health:** POSITIVE (+) — Plants with access to soil moisture stay green, hydrated, and alive. When soil moisture drops below a critical threshold, plants wilt, dry out, and become dead fuel for fire.
- **Vegetation Health to Fire Spread Rate:** NEGATIVE (-) — Healthy, hydrated vegetation resists ignition and slows fire spread. Green plants contain 100-200% of their dry weight in water, making them extremely difficult to burn. Dry, dead vegetation burns rapidly.

Key Discoveries

- Beaver dams raise the groundwater level by spreading and storing water across the floodplain — creating an underground reservoir
- Higher groundwater keeps soil moist even during drought, allowing vegetation to stay green and hydrated
- Green, hydrated vegetation does NOT burn easily — it creates a natural firebreak that slows or stops fire spread
- Without beaver dams, drought quickly drains soil moisture, vegetation dries out, and fire spreads rapidly through the landscape
- Beaver dams also REDUCE surface runoff — keeping more water in the landscape instead of letting it flow away

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Beavers prevent all wildfires"	Beavers do not prevent wildfire — they create LOCALIZED fire-resistant corridors along waterways. The green corridor may be only 50-200 meters wide on either side of the stream. Upland areas away from the stream still burn. Beavers are part of a wildfire resilience strategy, not a complete solution.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Water on the surface is what stops fire"	Surface water (ponds, streams) helps, but the REAL mechanism is UNDERGROUND water. Beaver dams raise the water table, which keeps soil saturated and plant roots hydrated. A dry hillside with a stream at the bottom can still burn. But a hillside with a HIGH WATER TABLE — where the underground water reaches plant roots — stays green and resists fire.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Beaver dams last forever once built"	Beaver dams require constant maintenance by the beaver colony. Without active beavers repairing damage from storms, ice, and water pressure, dams begin failing within months. Once the dam fails, stored groundwater drains away, soil dries out, vegetation health declines, and the fire resistance disappears. This is why beaver POPULATIONS matter — not just individual dams.